

# Øvingsoppgave 9

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$$y_n - 5y_{n-1} + 6y_{n-2} = 5 \cdot 4^n$$

$$y_n = y_n^{(h)} + y_n^{(p)}$$

6.3.8 a)

TML:

$$y_n - 5y_{n-1} + 6y_{n-2} = 0 \quad | y_n = A\lambda^n$$

$$A\lambda^n - 5A\lambda^{n-1} + 6A\lambda^{n-2} = 0 \quad | : (A\lambda^{n-2})$$

$$\lambda^2 - 5\lambda + 6 = 0 \quad | : abc$$

$$\begin{aligned} n - (n-2) &= n - n + 2 = 2 \\ n-1 - (n-2) &= n-1 - n + 2 = 1 \\ n-2 - (n-2) &= n-2 - n + 2 = 0 \end{aligned}$$

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\lambda_{1,2} = \frac{-(-5) \pm \sqrt{(-5)^2 - 4 \cdot 1 \cdot 6}}{2 \cdot 1}$$

$$\lambda_{1,2} = \frac{5 \pm 1}{2} \quad \lambda_1 = 3 \wedge \lambda_2 = 2$$

$$y_n^{(h)} = A \cdot 3^n + B \cdot 2^n \quad \text{TML}$$

$$y_n^{(p)} = 5 \cdot 4^n \quad | 5 \Rightarrow K$$

$$(K \cdot 4^n) - 5(K \cdot 4^{n-1}) + 6(K \cdot 4^{n-2}) = 5 \cdot 4^n$$

$$K \cdot 4^n - 5K \cdot 4 \cdot 4^{n-2} + 6K \cdot 4^{n-2} = 5 \cdot 4^n \quad | : 4^{n-2} \quad n - (n-2) = 2$$

$$K \cdot 16 - 20K + 6K = 5 \cdot 16$$

$$2K = 80 \quad | : 2$$

$$K = 40$$

$$y_n^{(p)} = 40 \cdot 4^n$$

$$y_n = y_n^{(h)} + y_n^{(p)} = \underline{\underline{A \cdot 3^n + B \cdot 2^n + 40 \cdot 4^n}}$$

6.3.8

d)

$$y_n - 2y_{n-1} + y_{n-2} = 6n$$

$$\text{THL: } y_n - 2y_{n-1} + y_{n-2} = 0 \quad | y_n = A \cdot \lambda^n$$

$$\lambda^2 - 2\lambda + 1 = 0 \quad | \text{abc}$$

$$\lambda = \frac{-(-2) \pm \sqrt{(-2)^2 - 4 \cdot 1 \cdot 1}}{2 \cdot 1}$$

$$\lambda = \frac{2 \pm 0}{2} = 1$$

$$y_n = A \cdot 1^n + Bn \cdot 1^n$$

$$\underline{\underline{y_n = A + Bn}}$$

Opag 1

$$-3x_1 - 2x_2 = -5$$

$$4x_1 + x_2 = 10$$

$$\overline{A} = \begin{bmatrix} -3 & -2 & -5 \\ 4 & 1 & 10 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & -1 & 5 \\ 4 & 1 & 10 \end{bmatrix} R'_1 = R_1 + R_2$$

$$\sim \begin{bmatrix} 1 & -1 & 5 \\ 0 & 5 & -10 \end{bmatrix} R'_2 = R_2 - 4R_1$$

$$\sim \begin{bmatrix} 1 & -1 & 5 \\ 0 & 1 & -2 \end{bmatrix} R'_2 = R_2 : 5$$

$$\begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & -2 \end{bmatrix} R'_1 = R_1 + R_2$$

$$\underline{\underline{x_1 = 3 \quad x_2 = -2}}$$

oppg 2

$$x_1 + x_2 + 5x_3 = 6$$

$$2x_1 - 3x_2 + 4x_3 = 16$$

$$4x_1 + 2x_2 - 5x_3 = 3$$

$$T = \begin{bmatrix} 1 & 1 & 5 & 6 \\ 2 & -3 & 4 & 16 \\ 4 & 2 & -5 & 3 \end{bmatrix} \begin{matrix} R_1 \\ R_2 \\ R_3 \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 1 & 5 & 6 \\ 0 & -5 & -6 & 4 \\ 4 & 2 & -5 & 3 \end{bmatrix} \begin{matrix} \\ R'_2 = R_2 - 2R_1 \\ \\ \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 1 & 5 & 6 \\ 0 & -5 & -6 & 4 \\ 0 & -2 & -25 & -21 \end{bmatrix} \begin{matrix} \\ \\ R'_3 = R_3 - 4R_1 \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 1 & 5 & 6 \\ 0 & 1 & 69 & 67 \\ 0 & -2 & -25 & -21 \end{bmatrix} \begin{matrix} \\ R'_2 = R_2 - 3R_3 \\ \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 1 & 5 & 6 \\ 0 & 1 & 69 & 67 \\ 0 & 0 & 113 & 113 \end{bmatrix} \begin{matrix} \\ \\ R'_3 = R_3 + 2R_2 \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 1 & 5 & 6 \\ 0 & 1 & 69 & 67 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{matrix} \\ \\ R'_3 = R_3 : 113 \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{matrix} R'_1 = R_1 - 5R_3 \\ R'_2 = R_2 - 69R_3 \\ \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{matrix} R'_1 = R_1 - R_2 \\ \\ \end{matrix}$$

$$\underline{\underline{x_1 = 3 \quad x_2 = -2 \quad x_3 = 1}}$$

oppg 3

$$x_1 + x_2 + 2x_3 = 7$$

$$2x_1 + 4x_2 - 3x_3 = -9$$

$$3x_1 + 6x_2 - 5x_3 = -15$$

$$T = \begin{bmatrix} 1 & 1 & 2 & 7 \\ 2 & 4 & -3 & -9 \\ 3 & 6 & -5 & -15 \end{bmatrix} \begin{matrix} R_1 \\ R_2 \\ R_3 \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 1 & 2 & 7 \\ 0 & 2 & -7 & -23 \\ 0 & 3 & -11 & -36 \end{bmatrix} \begin{matrix} R'_1 = R_1 - 2R_2 \\ R'_2 = R_2 - 2R_1 \\ R'_3 = R_3 - 3R_1 \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \end{bmatrix} \begin{matrix} R'_1 = R'_1 - 2R'_2 \\ R'_2 = R'_2 + 3R'_3 \\ \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 1 & 2 & 7 \\ 0 & 1 & -3 & -10 \\ 0 & 3 & -11 & -36 \end{bmatrix} \begin{matrix} R'_2 = 2R'_2 - R'_3 \\ \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \end{bmatrix} \begin{matrix} R'_1 = R'_1 - R'_2 \\ \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 1 & 2 & 7 \\ 0 & 1 & -3 & -10 \\ 0 & 0 & -2 & -6 \end{bmatrix} \begin{matrix} R'_3 = R'_3 - 3R'_2 \\ \end{matrix}$$

$$\underline{\underline{x_1 = 2 \wedge x_2 = -1 \wedge x_3 = 3}}$$

$$\sim \begin{bmatrix} 1 & 1 & 2 & 7 \\ 0 & 1 & -3 & -10 \\ 0 & 0 & 1 & 3 \end{bmatrix} \begin{matrix} R'_3 = R'_3 : -2 \\ \end{matrix}$$

oppg 4

$$-3x_1 + x_2 - x_3 = 12$$

$$2x_1 + 3x_3 = -9$$

$$2x_1 - 3x_2 + x_3 = -13$$

$$T = \begin{bmatrix} -3 & 1 & -1 & 12 \\ 2 & 0 & 3 & -9 \\ 2 & -3 & 1 & -13 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 3 & -9 \\ 2 & -3 & 1 & -13 \\ -3 & 1 & -1 & 12 \end{bmatrix} \begin{matrix} R_1 \\ R_2 \\ R_3 \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 1 & 5 & -6 \\ 2 & -3 & 1 & -13 \\ -3 & 1 & -1 & 12 \end{bmatrix} \begin{matrix} R'_1 = 2R_1 + R_3 \\ \\ \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 0 & 44 & -47 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & -1 \end{bmatrix} \begin{matrix} R'_1 = R_1 - R_2 \\ R'_2 = R_2 + 39R_3 \\ \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 1 & 5 & -6 \\ 0 & -5 & -9 & -1 \\ 0 & 4 & 14 & -6 \end{bmatrix} \begin{matrix} R'_2 = R_2 - 2R_1 \\ R'_3 = R_3 + 5R_2 \\ \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & -5 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & -1 \end{bmatrix} \begin{matrix} R'_1 = R_1 - 44R_3 \\ \\ \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 1 & 5 & -6 \\ 0 & 25 & 45 & 5 \\ 0 & 4 & 14 & -6 \end{bmatrix} \begin{matrix} R'_2 = R_2 \cdot (-5) \\ \\ \end{matrix}$$

$$\underline{x_1 = -3 \quad x_2 = 2 \quad x_3 = -1}$$

$$\sim \begin{bmatrix} 1 & 1 & 5 & -6 \\ 0 & 1 & -39 & 41 \\ 0 & 4 & 14 & -6 \end{bmatrix} \begin{matrix} R'_2 = R_2 - 6R_3 \\ \\ \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 1 & 5 & -6 \\ 0 & 1 & -39 & 41 \\ 0 & 0 & 170 & -170 \end{bmatrix} \begin{matrix} R'_3 = R_3 - 4R_2 \\ \\ \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 1 & 5 & -6 \\ 0 & 1 & -39 & 41 \\ 0 & 0 & 1 & -1 \end{bmatrix} \begin{matrix} R'_3 = R_3 : 170 \\ \\ \end{matrix}$$

10.1.6

$$a) \quad 2x_1 + 5x_2 = 0$$

$$3x_1 - 2x_2 = 0$$

$$T \begin{bmatrix} 2 & 5 & 0 \\ 3 & -2 & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} 4 & 10 & 0 \\ 3 & -2 & 0 \end{bmatrix} R'_1 = R_1 + 2$$

$$\begin{bmatrix} 1 & 12 & 0 \\ 3 & -2 & 0 \end{bmatrix} R'_1 = R_1 - R_2$$

$$\begin{bmatrix} 1 & 12 & 0 \\ 0 & -38 & 0 \end{bmatrix} R'_2 = R_2 - 3R_1$$

$$\begin{bmatrix} 1 & 12 & 0 \\ 0 & 1 & 0 \end{bmatrix} R'_2 = R_2 : -38$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} R'_1 = R_1 - 12R_2$$

$$\underline{\underline{x_1 = 0 \quad \wedge \quad x_2 = 0}}$$

10.1.6

d)

$$3x_1 + 2x_2 - 5x_3 = 0$$

$$2x_1 - x_3 = 0$$

$$5x_1 - 2x_2 + x_3 = 0$$

$$T = \begin{bmatrix} 3 & 2 & -5 & 0 \\ 2 & 0 & -1 & 0 \\ 5 & -2 & 1 & 0 \end{bmatrix} \begin{matrix} R_1 \\ R_2 \\ R_3 \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 2 & -4 & 0 \\ 2 & 0 & -1 & 0 \\ 5 & -2 & 1 & 0 \end{bmatrix} \begin{matrix} R'_1 = R_1 - R_2 \\ \\ \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 2 & -4 & 0 \\ 0 & -4 & 7 & 0 \\ 0 & -12 & 21 & 0 \end{bmatrix} \begin{matrix} R'_2 = R_2 - 2R_1 \\ R'_3 = R_3 - 5R_1 \\ \end{matrix}$$

$$\begin{bmatrix} 1 & 2 & -4 & 0 \\ 0 & 1 & -\frac{7}{4} & 0 \\ 0 & 1 & -\frac{7}{4} & 0 \end{bmatrix} \begin{matrix} R'_2 = R_2 : (-4) \\ R'_3 = R_3 : (-12) \\ \end{matrix}$$

$$\begin{bmatrix} 1 & 2 & -4 & 0 \\ 0 & 1 & -\frac{7}{4} & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} R'_3 = R_3 - R_2 \\ \\ \end{matrix}$$

$$x_3 = t, t \in \mathbb{R}$$

$$R_3: x_3 = t, t \in \mathbb{R}$$

$$x_1 = \frac{1}{2}t \wedge x_2 = \frac{7}{4}t \wedge x_3 = t$$

$$R_2: x_2 - \frac{7}{4}x_3 = 0 \wedge x_2 =$$

$$x_2 = \frac{7}{4}x_3 = \frac{7}{4}t$$

$$R_1: x_1 + 2x_2 - 4x_3 = 0$$

$$x_1 + 2 \cdot \frac{7}{4}t - 4t = 0$$

$$x_1 + \frac{7}{2}t - 4t = 0$$

$$x_1 - \frac{1}{2}t = 0$$

$$x_1 = \frac{1}{2}t$$